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Original Article

Domain-adapted foundation model for automated cardiac CT substructure segmentation for thoracic radiotherapy

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ABSTRACT

Purpose: Radiation-induced cardiotoxicity contributes to non-cancer mortality in patients treated with thoracic radiotherapy. Radiation dose to specific cardiac substructures is more predictive of adverse cardiac outcomes than whole-heart dose. This study aimed to develop a robust, automated cardiac substructure segmentation framework tailored to thoracic RT imaging.

Methods and materials: We curated CardiacSubstructSeg, a contrast-enhanced RT simulation CT dataset comprising 69 patients with left-sided lung cancer and expert manual annotations of 21 cardiac substructures. We proposed DINO-CardiacSeg, a novel segmentation framework built upon a CT domain-adaptive self-supervised pretraining. Model performance was assessed using five-fold cross-validation and compared with state-of-the-art methods. Transferability was assessed via fine-tuning and evaluation on the TotalSegmentator cardiac subset (n = 603).

Results: On CardiacSubstructSeg, DINO-CardiacSeg achieved the highest overall performance across all metrics (65.95% ± 7.91% DSC and 7.91 ± 4.40 mm HD95). Performance gains were consistent across small and low-contrast structures. In transfer-learning evaluation on TotalSegmentator, DINO-CardiacSeg maintained superior performance (DSC: 90.81% ± 8.23%, HD95: 6.07 ± 13.17 mm), demonstrating effective adaptation after fine-tuning to a dataset with differing annotation protocols and imaging characteristics.

Conclusions: DINO-CardiacSeg enables accurate and scalable segmentation of clinically relevant cardiac substructures on thoracic RT simulation CT through CT-specific foundation model pretraining. This approach outperforms existing methods on internal evaluation and fine-tuning to a public cardiac CT dataset, supporting its potential for substructure-level cardiac dosimetry and improved cardiotoxicity risk stratification in lung radiotherapy.

Introduction

Despite a global decrease in cancer mortality rates driven by reduced smoking habits and improved screening measures, lung and bronchus cancers account for 20.2% of all cancer-related deaths, the highest mortality rate of all cancers, and comprise nearly half of all incident cancer cases in both men and women [1]. As more than 50% of patients with lung cancer receive radiation therapy as a component of their definitive treatment, there is continued interest in the need to reduce adverse side effects that contribute to treatment-related deaths [2].

As patients treated for lung cancer are surviving longer due to improvements in treatment strategy and technology, later-arising treatment-associated adverse cardiac effects are being observed with increased frequency [3]. Although advancements in external beam radiation therapy have reduced high-dose radiation exposure to nearby organs-at-risk, modern planning and delivery techniques such as

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intensity-modulated radiation therapy (IMRT) and volumetric-modulated arc therapy (VMAT) can increase the volume of surrounding tissue receiving low-dose radiation [4,5]. Due to the anatomical proximity of the heart, patients treated for primary tumors localized in the left lung are particularly vulnerable to increased cardiac radiation exposure during treatment, and therefore at increased risk for cardiotoxicity. Post-treatment cardiotoxicity contributes to non-cancer mortality in patients treated for lung and bronchus cancer, with Yi et al. demonstrating that radiotherapy (RT) was independently associated with worse cardiovascular-specific survival (HR = 1.18, $P < 0.001$) and greater deaths from heart disease in patients receiving radiation versus those not [6]. Emerging evidence underscores the critical need to strengthen cardiac protection in patients undergoing lung RT, thus necessitating the accurate prediction and prevention of cardiotoxicity as essential to improving long-term survivorship.

Traditionally, thoracic RT planning involves whole-heart contouring as a single organ-at-risk (OAR) on computed tomography (CT) images to simulate, calculate, and minimize radiation exposure as a cost-function of treatment planning. However, emerging evidence indicates that radiation dose to specific cardiac substructures is a more accurate predictor of treatment-associated cardiotoxicity than conventional whole-heart dose [7,8]. Landmark studies have demonstrated that radiation doses to specific cardiac substructures are strongly associated with treatment-related cardiac events following lung RT, while highlighting the limited predictive value of mean whole-heart dose as an OAR metric [9–11]. As the heart functions as a mixed serial-parallel organ, radiation injury to critical substructures, such as the coronary arteries or heart valves, can compromise whole-heart health [12–14]. Current OAR conventions of utilizing whole-heart dose may therefore underestimate cardiac risk by not accounting for heterogeneous dose exposure to substructures that may impart heightened cardiotoxicity risk.

Manual contouring of the cardiac substructures is labor-intensive, time-consuming, and subject to inter-observer variability, rendering the task difficult to incorporate into clinical practice. Small substructures, such as the coronary arteries, or those entirely non-visible on CT scans, such as the conduction nodes, are extremely challenging to delineate on simulation CT images, even with contrast enhancement. Patient-to-patient anatomic variability, as well as cardiac and respiratory motion interplay during imaging, obscure the visual quality of substructures and therefore decrease accuracy of contours. Hence, automated deep learning segmentation tools may facilitate consistent, large-scale cardiac contouring to clinically incorporate and personalize thoracic RT planning.

Previous deep learning models for cardiac substructure segmentation were trained on modest sample sizes and limited substructures. Moreover, existing open-source datasets have been developed on, and for, cardiac imaging purposes, limiting generalizability to lower-quality thoracic RT simulation images and associated models. To address these limitations, we present a large, expertly annotated dataset for cardiac substructure segmentation in thoracic radiotherapy, comprising contrast-enhanced CT simulation scans from 69 patients with left-sided lung cancer and retrospective manual annotations of 21 cardiac substructures.

We propose DINO-CardiacSeg, a novel cardiac substructure segmentation framework that builds on the DINOv3 vision foundation model [15]. Although DINOv3 has demonstrated strong performance in natural-image and medical-image segmentation tasks, its direct application to CT is challenged by substantial domain gaps, including imaging intensity characteristics and the need to encode anatomically meaningful information. To address these limitations, we developed a domain-adapted foundation model workflow that uses large-scale self-supervised learning to learn CT-specific representations. We curated a large, diverse CT dataset from four public sources and performed domain-adaptive pretraining to tailor DINOv3 foundation model to the CT imaging domain. Building on this pretrained model, we introduced a novel segmentation architecture designed for high-resolution cardiac substructure delineation. The proposed model incorporates (1) a DINOv3 adapter module that injects domain-adapted multi-scale features via deformable attention and (2) a hierarchical deformable attention decoder optimized to capture fine-grained anatomic boundaries while preserving global cardiac context.

DINO-CardiacSeg outperformed multiple state-of-the-art 2D segmentation methods (nnU-Net [16], U-Net++ [17], Swin-UNET [18], and Dino-UNet [19]) on our institutional cardiac substructure dataset and, after fine-tuning, maintained superior segmentation accuracy on the public TotalSegmentator cardiac subset [20]. By enabling precise substructure-level dose assessment, this approach supports patient-specific radiotherapy planning and advances precision cardiac dosimetry to improve the prediction and prevention of radiation-induced cardiotoxicity.

Materials & methods

Dataset characteristics

Sixty-nine patients with left-sided lung cancer (36 male, 33 female) treated at a single institution with photon radiation therapy (RT) between 2015 and 2025 were identified from institutional electronic health records (Supplementary Materials Table A.1). The cohort represents patients diagnosed with both non-small cell and small cell lung cancers. Eligibility criteria included the availability of a contrast-enhanced CT simulation scan within 1 month prior to treatment start. All patients received the contrast agent Omnipaque 350 intravenously prior to acquisition of simulation images. Images were acquired using Siemens CT scanners with in-plane resolution of 0.71–1.36 mm and slice thickness of 2–5 mm, tube voltage of 100–120 kVp, and tube current ranging from 91–645 mA. Patient-protected health information (PHI) metadata was removed from the DICOM files prior to model training and testing.

Substructure annotations

For each patient, 21 cardiac substructures were manually delineated on a contrast-enhanced CT scan by a trained annotator. These included the heart chambers (right and left atria, right and left ventricles), valves (mitral, tricuspid, aortic, and pulmonary), conduction nodes (sinoatrial and atrioventricular), major vessels and related structures (aortic arch, ascending and descending aorta, superior and inferior vena cava, pulmonary artery and veins), and four coronary arteries (right and left coronary arteries, left anterior descending, and left circumflex). All segmentations were performed in Velocity 4.1 (Varian Medical Systems).

To promote institutional consistency and minimize atlas-related variability, contours were generated through an integrated approach combining multiple gold-standard references. The RTOG “Atlases for Organs at Risk (OARs) in Thoracic Radiation Therapy” and Feng et al. atlas served as primary visual guides for the chambers, great vessels, and coronary arteries [21,22]. Heart valves were contoured following the Socha et al. atlas, while delineation of the conduction nodes was guided by Loap et al. [23,24]. The cranial and caudal boundaries of each substructure were further informed by Vaugier et al. and Milo et al. [25,26]. A consolidated institutional framework was then developed to harmonize these references and account for patient-specific anatomical variability. Comprehensive written contouring guidelines for all 21 substructures are available upon request. All contours were reviewed by a board-certified radiation oncologist and two board-certified medical physicists to ensure clinical accuracy.

Domain-adaptive self-supervised pretraining

While vision foundation models pretrained on large datasets have demonstrated remarkable performance in natural images, directly

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adapting them for CT images remains non-trivial, due to considerable gaps in imaging characteristics. To address this challenge, we propose to perform large-scale domain-adaptive pretraining on CT images using DINOv3 (Fig. 1a), a state-of-the-art SSL method known for learning superior dense features [15]. First, we curated a large-scale unlabeled CT dataset with an emphasis on thoracic region. Our pretraining dataset included AbdomenAtlas [27], Totalsegmentator V2 [20], LIDC [28], and StonyBrookChestCT [29]. This data curation provides broad anatomical coverage (over 100 structures) across abdominal, thoracic, and pelvic regions, ensuring both scale and heterogeneity for domain-adaptive pretraining. All 3D volumes were resampled to an in-plane spacing of 0.75 mm and 0.75 mm and an axial spacing of 1.5 mm. Only axial slices were utilized as model inputs, resampled to a uniform resolution of 256 × 256. The final pretraining dataset included 4,440,784 axial slices extracted from 13,799 3D volumes.

For the encoder architecture, we employed the standard base Vision Transformer (ViT-B), consisting of approximately 86 million parameters. The pretraining process was divided into two stages. In stage 1, we optimized a combined objective comprising the self-distillation loss L_{DINO} and the masked image modeling loss L_{iBOT}. At the start of training, a teacher–student network pair was initialized. For each input image, the teacher network received a global crop, while the student network received eight local crops generated through random cropping. We enforced the CLS token of each local crop to match the CLS token of the teacher's global crop, encouraging the model to learn scale-consistent visual representations. The resulting embeddings were passed through an MLP projection head to obtain prototype scores, followed by softmax normalization and cross-entropy loss. For the masked image modeling component, the objective was to strengthen spatial understanding by requiring the model to recover the semantics of masked patches. A masked input image was passed through the student network, and its prototype scores were compared with the teacher's scores at the corresponding unmasked locations. Additionally, a prototype regularization loss L_{Koleo} was applied to promote a uniform distribution of feature prototypes.

$$\mathscr{L}_{\text{Stage1}} = \mathscr{L}_{\text{DINO}} + \mathscr{L}_{\text{iBOT}} + 0.1 \cdot \mathscr{L}_{\text{Koleo}}.$$

In stage 2, we adapted the SSL framework for higher-resolution training, motivated by the relatively small volume and fine anatomical detail of cardiac substructures such as the coronary arteries. To better preserve these structures, we increased the input image resolution to 512 × 512 to continue pretraining. However, simple upscaling to higher resolution may degrade local feature quality, as global and local SSL objectives are only weakly correlated. To mitigate this limitation, we incorporated Gram anchoring. This approach regularizes the Gram matrix, which is the matrix of all pairwise dot products of patch features within an image. Specifically, we encouraged the Gram matrix of the student network to remain aligned with that of an earlier teacher checkpoint, referred to as the Gram teacher, which retained stronger dense (local) features from early training. Formally, given an image with P patches and a network that operates in dimension d, let X_S and X_G denote the $P \times d$ matrix of L_2-normalized local features of the student and the Gram teacher, respectively. We defined the loss $\mathscr{L}_{\text{Gram}}$ as follows:

$$\mathscr{L}_{\text{Gram}} = \|X_S \cdot X_S^{\top} - X_G \cdot X_G^{\top}\|_F^2.$$

This loss was only computed on the global crops across all patch tokens. In our implementation, we started this stage after 100k iterations of pretraining, for a total of 50k iterations. The stage 2 loss was defined as:

$$\mathscr{L}_{\text{Stage2}} = \mathscr{L}_{\text{DINO}} + \mathscr{L}_{\text{iBOT}} + \mathscr{L}_{\text{Koleo}} + 2^{*}\mathscr{L}_{\text{Gram}}.$$

DINO-CardiacSeg architecture

After domain-adaptive pretraining, we developed DINO-CardiacSeg, a segmentation model tailored to cardiac substructure segmentation, leveraging the CT-adapted DINOv3 encoder (Fig. 1b). DINO-CardiacSeg was composed of three main modules:

(1) CT-adapted ViT encoder. The input CT slice was first processed by the pretrained ViT-Base encoder, which embedded the image into a sequence of 16 × 16 non-overlapping patches and produces transformer feature representations adapted to the CT domain.
(2) Hybrid CNN–Transformer adapter. In parallel, the input image was passed through a lightweight convolutional stem to generate

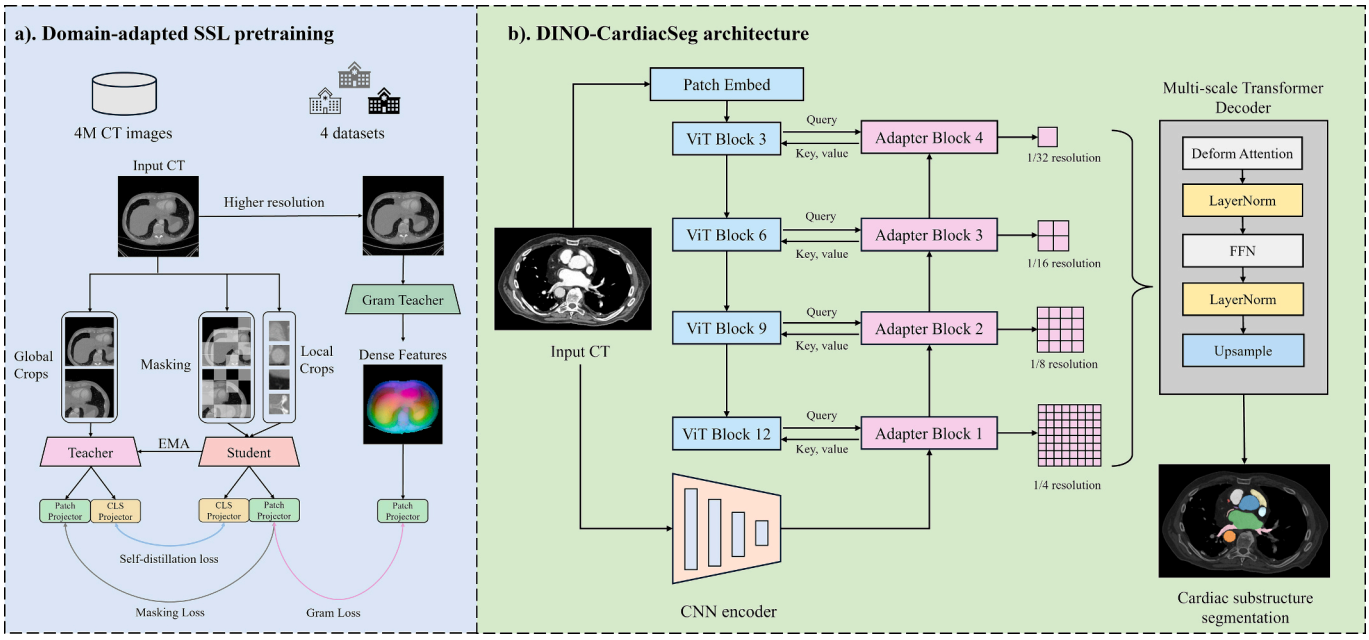


Fig. 1. (a) Domain-adapted self-supervised pretraining of a DINOv3 vision transformer on large-scale unlabeled CT data using self-distillation, masked image modeling, and Gram anchoring to learn CT-specific dense representations. (b) DINO-CardiacSeg architecture for cardiac substructure segmentation on radiotherapy simulation CT, consisting of a CT-adapted ViT encoder, a hybrid CNN–Transformer adapter with multi-scale cross-attention, and a multi-scale deformable attention decoder to produce high-resolution cardiac substructure segmentations.

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multi-scale feature maps at 1/8, 1/16, and 1/32 of the input resolution. The transformer encoder layers of ViT were divided evenly into four blocks (each containing three layers). For each block, we introduced two cross-attention interactions between the ViT features and the convolutional features. In the first cross-attention layer, the flattened CNN feature maps were used as keys and values, while the ViT block output served as queries. In the second cross-attention layer, the queries came from the updated features of the previous layer, and the keys and values came from the ViT block output. These bidirectional interactions allowed the adapter to fuse global ViT context with local convolutional detail, producing high-quality multi-scale representations. After cross-attention fusion, the resulting features were reshaped into spatial maps at 1/4, 1/8, 1/16, and 1/32 resolution.

(3) (3). Multi-scale deformable attention decoder. With the feature pyramids, the decoder leveraged deformable attention to aggregate cross-scale context. It first projected each feature map to a common channel dimension using 1×1 convolutions followed by group normalization. Next, the features were passed to a multi-scale deformable transformer encoder, which aggregated information across spatial locations and feature levels using deformable attention. Lastly, the feature maps were progressively upsampled via lateral 1×1 convolutions and 3×3 convolutions to obtain high-resolution predictions for cardiac substructures.

Implementation

Both pretraining and finetuning were performed on a NVIDIA DGX A100 cluster. Pretraining used two A100 GPUs (80G) for approximately two days. We pretrained for a total of 120k iterations, using a global batch size of 512. For stage 1, we used a learning rate of 2e-4 and trained for 100k iterations. For stage 2, an exponential moving average (EMA) model at 20k iterations was selected as the Gram teacher, and pretraining was continued using the gram anchoring objective for an additional 20k iterations. Finetuning experiments were performed within the nnU-Net framework [16] in PyTorch. All models were trained for 1,000 epochs, with each epoch consisting of 250 training steps. For the CardiacSubstructSeg dataset, CT images were resampled to an in-plane resolution of 0.97 mm while retaining the original axial spacing, and a patch size of 512×512 was used. For the TotalSegmentator cardiac dataset, images were resampled to an in-plane resolution of 1.5 mm with a patch size of 256×256.

Baseline comparisons and evaluation metrics

We performed extensive comparisons between DINO-CardiacSeg and several state-of-the-art deep learning segmentation methods commonly used in medical imaging. The evaluated baseline models included: (1) nnU-Net [16], (2) U-Net++ [17], (3) Swin-UNET [18], and (4) DINO-UNet [19]. All models were trained and evaluated using five-fold cross-validation on the CardiacSubstructSeg dataset. For each fold, patient-level splits were used to ensure that scans from the same patient do not appear in both training and testing sets. Model performance was evaluated across all 21 annotated cardiac substructures, and metrics were computed on a per-structure basis and then averaged across structures and patients.

Segmentation accuracy was assessed using four standard metrics: Dice similarity coefficient (DSC), 95th percentile Hausdorff distance (HD95), surface dice coefficient (SDC), and average symmetric surface distance (ASSD). DSC and SDC quantify volumetric and surface overlap, respectively, while HD95 and ASSD evaluate boundary accuracy and robustness to outliers. Surface dice coefficient was computed on the predicted and reference segmentation surfaces in physical space. Specifically, let S_p and S_g denote the predicted and ground-truth surfaces, respectively, and let $d(x, S)$ denote the minimum Euclidean distance from surface point x to surface S. SDC at tolerance τ was defined as

$$\mathrm{SDC}_\tau = \frac{\left|\{x \in S_p : d(x, S_g) \le \tau\}\right| + \left|\{y \in S_g : d(y, S_p) \le \tau\}\right|}{|S_p| + |S_g|},$$

with $\tau = 2$ mm. All evaluation metrics were computed on 3D binary segmentation masks in physical millimeter space using the voxel spacing of each scan; for SDC, $|S_p|$ and $|S_g|$ denote the total number of surface elements in the discrete surface representation. Thus, a surface element on either segmentation was counted as correctly matched if its closest-point distance to the opposing surface was within 2 mm. For statistical comparison, the patient served as the unit of analysis. For each patient, structure-level metric values were averaged across all 21 substructures to yield a single patient-level summary score. Patient-level paired differences between DINO-CardiacSeg and nnU-Net were then computed, and statistical significance was assessed using a two-sided permutation test (50,000 sign permutations) on these paired differences, with the test statistic defined as the mean paired difference.

For the overall internal comparison (Supplementary Materials Table A.2), multiplicity across the four reported overall metrics (DSC, HD95, SDC, and ASSD) was controlled using Bonferroni correction with $m = 4$, corresponding to a corrected significance threshold of $\alpha = 0.0125$. For the organ-group analyses in Table 1, multiplicity across 10 comparisons (5 organ groups $\times$ 2 metrics) was controlled using Bonferroni correction with $m = 10$, corresponding to a corrected significance threshold of $\alpha = 0.005$. For the external overall comparison in Table 3, multiplicity across the same four overall metrics was likewise controlled using Bonferroni correction with $m = 4$ and $\alpha = 0.0125$. Additionally, 95% bootstrap confidence intervals (10,000 resamples) for the mean patient-level metric are reported in Table 1 to quantify uncertainty. Results were reported as mean $\pm$ standard deviation across all patients. Reported p-values are unadjusted two-sided permutation-test p-values. Statistical significance was determined using the Bonferroni-corrected α thresholds for each prespecified family of comparisons.

We defined a segmentation failure as an empty (null) prediction for a given structure in a given patient. In such cases, overlap metrics (DSC, SDC) were set to 0, and distance metrics (HD95, ASSD) were assigned a finite penalty equal to the maximum image diagonal of the scan, computed as the Euclidean distance across the full image volume. These failure-penalized values were used consistently for all patient-level summary statistics, hypothesis tests, and any main-text tables reporting these metrics. Per-substructure failure rates are reported separately in Supplementary Materials Table A.3 to provide a descriptive breakdown of failure frequency by method and structure. This protocol ensures that all patients contribute to summary statistics and statistical tests regardless of prediction quality, avoiding selection bias from case exclusion.

Results

As summarized in Table A.2 of the Supplementary Materials, DINO-CardiacSeg achieved the highest overall performance across all evaluation metrics on the CardiacSubstructSeg dataset. In the prespecified overall internal analysis, after Bonferroni correction for four overall metrics ($\alpha = 0.0125$), DINO-CardiacSeg demonstrated statistically significant improvement compared with nnU-Net in DSC ($p = 0.001$), SDC ($p = 0.0003$), and ASSD ($p = 0.008$). Although DINO-CardiacSeg also achieved a lower HD95 ($p = 0.024$), this difference did not meet the corrected significance threshold for the four-metric family. In contrast, transformer-based models without domain-adaptive pretraining (Swin-UNET and DINO-UNet) exhibited substantially inferior performance, particularly in surface-based metrics, suggesting limited robustness for fine-grained cardiac anatomy in thoracic CT images.

Organ-level analysis further highlighted the strengths of DINO-CardiacSeg across both large and small cardiac structures (Table 1). For this separate organ-group family of tests, Bonferroni correction was applied across 10 comparisons ($\alpha = 0.005$). For cardiac chambers and

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Table 1
Organ-level comparison of our method and existing segmentation methods on CardiacSubstructSeg dataset (n = 69). Data were denoted as mean [95% CI]. P-values were obtained by comparing DINO-CardiacSeg with nnU-Net using patient-level paired permutation tests. For organ-group analyses, Bonferroni correction was applied across 10 comparisons (5 organ groups × 2 metrics: DSC and HD95), corresponding to a corrected significance threshold of $p < 0.005$. **Bold** indicates best result. * $p < 0.005$. Cases with segmentation failure have overlap metrics set to 0 and distance metrics set to the maximum image diagonal of the scan.

Methods	Chambers		Great Vessels		Coronary Arteries		Conduction Nodes		Valves	
	DSC (%)	HD95 (mm)	DSC (%)	HD95 (mm)	DSC (%)	HD95 (mm)	DSC (%)	HD95 (mm)	DSC (%)	HD95 (mm)
nnUNet	86.35 [84.78, 87.92]	8.50 [7.32, 9.68]	86.30 [84.44, 88.16]	7.41 [4.22, 10.60]	38.64 [33.44, 43.84]	14.63 [11.82, 17.44]	29.91 [25.25, 34.57]	7.76 [6.60, 8.92]	44.45 [38.52, 50.38]	8.88 [7.47, 10.29]
UNet ++	86.77 [85.33, 88.21]	9.26 [7.25, 11.27]	85.80 [83.69, 87.91]	8.78 [5.29, 12.27]	**40.41 [35.21, 45.61]**	13.22 [11.07, 15.37]	28.05 [24.28, 31.82]	8.11 [6.96, 9.26]	40.88 [34.99, 46.77]	10.16 [8.16, 12.16]
SwinUNET	83.92 [82.55, 85.29]	16.08 [11.64, 20.52]	78.02 [74.79, 81.25]	12.76 [10.54, 14.98]	12.19 [8.18, 16.20]	17.44 [14.73, 20.15]	0 [0,0]	833.96 [779.02, 891.39]	25.70 [19.83, 31.57]	11.58 [10.16, 13.00]
DINO-UNet	83.15 [81.30, 85.00]	15.32 [11.72, 18.92]	80.61 [77.99, 83.23]	16.24 [9.47, 23.01]	0 [0,0]	833.96 [779.02, 891.39]	9.78 [6.94, 12.62]	17.82 [13.77, 21.87]	39.25 [34.09, 44.41]	12.06 [9.83, 14.29]
DINO-CardiacSeg (Ours)	**88.60*** **[87.54, 89.66]**	**7.25*** **[6.26, 8.24]**	**86.56** **[84.75, 88.37]**	**6.60*** **[4.98, 8.22]**	39.91 [35.25, 44.57]	**7.76* [6.60, 8.92]**	**30.85** **[26.99, 34.71]**	**7.36* [6.79, 7.93]**	**46.58*** **[40.43, 52.73]**	**8.66** **[7.28, 10.04]**

great vessels, DINO-CardiacSeg demonstrated the highest mean DSC and the lowest HD95, achieving statistically significant improvements in chamber segmentation compared with nnU-Net ($p = 0.0001$ for DSC, $p = 0.001$ for HD95). Performance gains were also significant for HD95 in small substructures, including the coronary arteries and cardiac valves. Notably, transformer-based baselines without CT-specific adaptation frequently failed to segment conduction nodes and coronary arteries, whereas DINO-CardiacSeg achieved stable performance. Vessel-level evaluation (Table 2) demonstrated that DINO-CardiacSeg achieved the lowest or near-lowest ASSD across all major vessels. Qualitative comparison in Fig. 2 also confirmed the segmentation improvements in substructures. Collectively, these results indicated that domain-adaptive foundation model pretraining combined with multi-scale deformable attention substantially improved segmentation robustness across a wide range of cardiac structures, especially for small, low-contrast, and geometrically complex anatomy relevant to cardiac dosimetry.

To evaluate transfer-learning performance, we fine-tuned all segmentation models pretrained on the CardiacSubstructSeg dataset on the TotalSegmentator cardiac subset. This public dataset comprised 603 CT scans derived from Totalsegmentator [20], with annotations for eight cardiac structures, including the aorta, inferior vena cava, pulmonary artery, and the four cardiac chambers (left and right atria and ventricles). All models were fine-tuned under identical training protocols and evaluated using five-fold cross-validation.

As summarized in Table 3, DINO-CardiacSeg consistently outperformed all baseline methods across all evaluation metrics. In the TotalSegmentator dataset, after Bonferroni correction for four overall metrics ($\alpha = 0.0125$), DINO-CardiacSeg achieved significantly higher mean DSC (90.81% ± 8.23%, $p = 1.91E-12$) and SDC (45.86% ± 12.44%, $p = 1.57E-29$), and significantly lower HD95 (6.07 mm ± 13.17 mm, $p = 0.0001$) and ASSD (1.53 mm ± 2.28 mm, $p = 1.15E-07$) than nnU-Net. Notably, while nnU-Net and U-Net++ achieved strong volumetric overlap on this dataset, their surface-based metrics remained comparatively lower, suggesting reduced sensitivity to fine boundary delineation. Transformer-based models without CT-specific domain adaptation (Swin-UNET and DINO-UNet) exhibited substantially degraded performance, particularly in HD95 and ASSD, suggesting reduced adaptability to the target dataset despite fine-tuning. In contrast, DINO-CardiacSeg maintained stable performance across both volumetric and surface metrics, demonstrating the benefit of domain-adaptive pretraining for fine-tuning-based transfer to new cardiac segmentation tasks on a distinct public dataset (Supplementary Materials Table A.4).

Table 3
Comparison of our method and existing segmentation methods on the TotalSegmentator Cardiac dataset segmenting 8 structures (n = 603). Data are denoted as mean ± SD. P-values were obtained by comparing with nnU-Net using patient-level paired permutation tests. **Bold** indicates best result. * $p < 0.0125$ with Bonferroni correction for 4 overall metrics.

Methods	DSC (%)	HD95 (mm)	SDC (%)	ASSD (mm)
nnUNet	89.32 ± 7.84	7.49 ± 15.77	40.99 ± 11.19	1.79 ± 2.71
UNet ++	89.12 ± 8.11	7.83 ± 14.79	41.52 ± 11.48	1.77 ± 2.19
SwinUNET	86.26 ± 11.58	11.83 ± 26.06	36.38 ± 11.40	2.59 ± 5.25
DINO-UNet	83.14 ± 10.79	24.48 ± 39.29	27.59 ± 8.54	4.41 ± 7.91
DINO-CardiacSeg (Ours)	**90.81 ± 8.23***	**6.07 ± 13.17***	**45.86 ± 12.44***	**1.53 ± 2.28***

Table 2
Great vessel segmentation results for our method and existing segmentation methods on CardiacSubstructSeg dataset (n = 69). Data were denoted as mean ± SD. **BOLD** indicates best result.

Methods	ASSD (mm)						
	Aortic Arch	Ascending Aorta	Descending Aorta	Pulmonary Artery	Pulmonary Veins	Superior Vena Cava	Inferior Vena Cava
nnUNet	1.33 ± 0.54	1.48 ± 1.46	0.96 ± 0.27	1.67 ± 0.77	**1.54 ± 0.56**	1.58 ± 0.48	**1.23 ± 0.55**
UNet ++	1.87 ± 1.43	1.42 ± 1.29	1.02 ± 0.44	1.43 ± 0.73	2.00 ± 0.99	1.57 ± 0.65	1.76 ± 1.74
SwinUNET	2.51 ± 0.86	2.23 ± 1.36	1.94 ± 1.49	3.84 ± 2.73	2.74 ± 1.70	2.66 ± 0.67	2.32 ± 0.96
DINO-UNet	2.53 ± 1.19	6.21 ± 8.31	2.21 ± 1.65	1.87 ± 0.85	2.39 ± 1.15	2.32 ± 0.51	2.04 ±1.29
DINO-CardiacSeg (Ours)	**1.23 ± 0.50**	**1.14 ± 0.45**	**0.87 ± 0.47**	**1.62 ± 0.79**	1.64 ± 0.58	**1.35 ± 0.33**	1.45 ± 0.64

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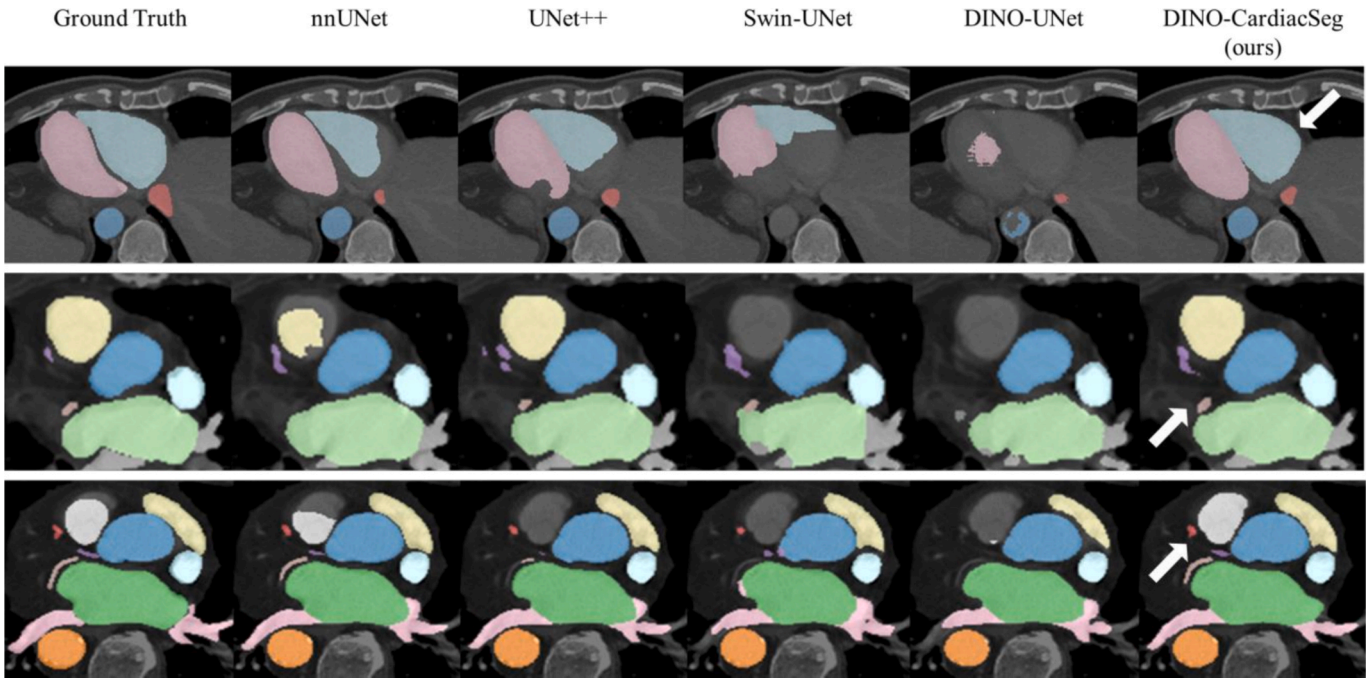


Fig. 2. Qualitative results comparing DINO-CardiacSeg with other segmentation methods on CardiacSubstructSeg dataset. White arrow indicates improvements.

Discussion

In this study, we introduce DINO-CardiacSeg, a CT-domain-adapted foundation model framework for the delineation of 21 cardiac substructures on contrast-enhanced RT simulation CT. By leveraging large-scale self-supervised pretraining on 13,799 CT volumes (4,440,784 axial slices) aggregated from four public datasets, the proposed approach addresses the domain mismatch between natural-image pretrained DINOv3 and clinical CT imaging. The incorporation of a DINOv3 adapter with deformable attention enables effective integration of CT-specific multi-scale representations, while the hierarchical deformable attention decoder facilitates precise delineation of small, low-contrast cardiac substructures while preserving global anatomic context. Additionally, we curated a detailed cardiac substructure segmentation dataset tailored to thoracic radiotherapy. Compared with prior studies that are characterized by smaller cohort sizes and narrowly defined cardiac anatomy, this study introduces a larger and more anatomically comprehensive dataset derived from contrast-enhanced lung radiotherapy simulation CTs (Table 4).

We benchmarked DINO-CardiacSeg and multiple state-of-the-art 2D segmentation models (nnU-Net [16], U-Net++ [17], Swin-UNET [18], and Dino-UNet [19]) using our cardiac substructure segmentation dataset aforementioned. Across all substructures, the proposed model achieved superior segmentation accuracy, demonstrating robust improvements over existing state-of-the-art methods. To further assess transferability, we evaluated fine-tuning performance on the publicly available TotalSegmentator cardiac subset [20]. DINO-CardiacSeg maintained the highest segmentation performance, indicating that CT domain-adapted pretraining facilitates effective adaptation after fine-tuning to a dataset with differing annotation protocols and imaging

Table 4
Comparison of our dataset, CardiacSubstructSeg, with those used in other studies proposing novel machine or deep learning models for CT-based cardiac substructure segmentation, excluding automatic atlas-based methods.

Study (Reference)	#Substructures	#Scans	Modality	Cancer Types Included	Model Type	Year
CardiacSubstructSeg (Ours)	21	69	Contrast CT	Left-lung cancer (NSCLC & SCLC)	DINOv3	2025
Chen *et al.* (Radiother Oncol 2024) [30]	19	100	Non-contrast CT	Lung cancer (NSCLC)	3D nnU-Net	2024
Chin *et al.* (Clin Oncol 2023) [31]	18	30	Non-contrast CT	Lung cancer (NSCLC)	Hybrid (DL + multi-atlas + geometric modeling)	2023
Finnegan *et al.* (Phys Eng Sci Med 2023) [32]	18	30	Non-contrast CT	Thoracic and breast cancer	DL + atlas mapping + geometric	2023
Momin *et al.* (Phys Med Biol 2022) [33]	15	55	Non-contrast CT	Thoracic (lymphoma, breast, lung, esophageal)	Retina U-Net with mutual enhancement	2022
Harms *et al.* (Med Phys 2021) [34]	15	55	Contrast (n=22) and Non-contrast CT (n=33)	Lung cancer (NSCLC & SCLC)	Mask-scoring R-CNN	2021
Morris *et al.* (Med Phys 2020) [35]	12	32	Non-contrast CT	Breast cancer (left-sided)	3D U-Net with CRF post-processing	2020
Wang *et al.* (Bioeng. 2023) [36]	10	60	Non-contrast CT	Cardiac patients, non-cancer	Coarse-to-fine cascaded network with anatomical priors	2023
Jin *et al.* (Med Phys 2021) [37]	9	129	Non-contrast CT	Breast cancer	3D CNN	2021
Walls *et al.* (Phys Imaging Radiat Oncol 2022) [38]	8	20	Non-contrast CT	Lung cancer	Pre-trained CNN	2022
Park & Chung (Comput Biol Med 2022) [39]	7	60	Non-contrast CT	Cardiac patients, non-cancer	CNN with shape-aware contour attention	2022

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characteristics.

In conclusion, DINO-CardiacSeg achieved the best overall performance on internal cross-validation and maintained superior accuracy in a transfer-learning evaluation on a distinct public dataset. These findings support the utility of CT domain-adapted pretraining for adaptation to new cardiac segmentation tasks, but do not constitute direct external validation without target-dataset adaptation. By enabling scalable and consistent substructure-level cardiac contouring, this work supports precision cardiac dosimetry and provides a practical foundation for improving prediction and prevention of radiation-associated cardiotoxicity in patients receiving thoracic radiotherapy.

Although our study provides important new insights and a further refined technologic approach, several limitations highlight areas for future work. First, the CardiacSubstructSeg cohort size remains modest (n=69) and originates from a single healthcare system, and acquisition was limited to contrast-enhanced simulation CT. Performance on non-contrast simulation CT and across multi-institutional scanners and protocols requires further validation. Second, substructures such as conduction nodes are not directly visible on CT and thus rely on atlas-guided inference; while this reflects clinical contouring practice, it introduces uncertainty in the "ground truth" and may cap achievable performance. Third, we evaluated segmentation quality using geometric metrics but did not directly quantify downstream clinical impact (e.g., changes in substructure dose metrics, plan quality, or toxicity prediction). While only objective assessments were included in this analysis, subsequent studies will include expert-completed subjective evaluations of the model-generated cardiac substructure OAR contours to assess their suitability for clinical treatment planning. Finally, our approach is currently slice-based; extending to 3D or 2.5D contexts may further improve continuity and reduce occasional discontinuities across slices for thin structures.

Our future work will investigate multi-institutional external validation on both contrast and non-contrast simulation CT, including patients treated with different techniques and motion-management strategies. Clinically, integrating DINO-CardiacSeg into a planning pipeline and evaluating effects on substructure dose–response modeling would provide direct evidence of utility. Methodologically, exploring uncertainty estimation and automatic quality assurance could improve safety for clinical deployment, particularly for small structures where occasional failure modes can disproportionately affect dosimetry. Finally, expanding annotations to additional substructures and standardizing contouring definitions across centers may facilitate harmonized studies and broader adoption.

CRediT authorship contribution statement

Yuheng Li: Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Vanessa L. Wildman:** Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Luke del Balzo:** Writing – review & editing, Investigation. **Richard Qiu:** . **Aparna H. Kesarwala:** . **Xiaofeng Yang:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.radonc.2026.111619.

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